

MA1521 L1 Week 01 Wed 20260812

Evaluate $\lim_{x \rightarrow \infty} \frac{x^2 + 2x + 3}{3x^2 - 2x + 4}$.

Evaluate $\frac{d}{dx} \left(\frac{x^4 + 1}{x} \right)$.

Evaluate $\frac{d}{dx} \ln(2 \cos x)$.

If $y = \frac{x}{x+5}$, what is $\frac{d^2y}{dx^2}$?

If $y = x^x$, what is $\frac{dy}{dx}$?

If $xy - \ln y = 1$ and $xy \neq 1$, what is $\frac{dy}{dx}$?

If $xy - \ln y = 1$ and $xy \neq 1$, what is $\frac{d^2y}{dx^2}$?

Determine the lines through the origin which are tangent to the graph of $y = 1 + (x - 1)^2$.

What is the equation for the tangent line to the graph of $y = x^3 - x^2 + 5x - 6$ at $x = 1$?

Evaluate $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{x - \frac{\pi}{2}} .$

Evaluate $\lim_{x \rightarrow 0} \frac{\cos x - 1 + \frac{x^2}{2}}{x^4} .$

What is the minimum value of $(x + 1)e^{-x}$ over all $x > 0$?

What is the maximum value of $(x + 1)e^{-x}$ over all $x > 0$?

Suppose the function f satisfies

$$f(x+y) = f(x) + f(y) + x^2y + xy^2 \quad \text{for all real } x, y, \quad \text{and also} \quad \lim_{x \rightarrow 0} \frac{f(x)}{x} = 1.$$

Determine $f'(x)$.

Give the answer as an **expression** in terms of x .